

centerline respaced to chords of $\ell = 5$ mm

θ_j = turn at vertex j , e.g. 0.876 rad (50.2°) gives $\theta_j/\ell = 0.175$ rad/mm



low

high

each token becomes 4 channels (32 in the model)

$\times W + b$



1

2

3

4

5

6

pad

pad

$z_j = (\theta_j/\ell - \mu)/\sigma$ with $\mu = 0.15$, $\sigma = 0.08$ over the training tokens; s_j = position along the vessel
the mean of the θ_j/ℓ row is $T_5 = 0.160$, the scalar the logistic baseline gets

dashed: padding, masked in attention